STAA57-Winter 2024: Worksheet-6

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Questions 1:

i. Download the "Pre-2023 Apartment Building Evaluation" data from the city of Toronto's open data portal.

https://open.toronto.ca/dataset/apartment-building-evaluation/

ii. Consider the variable "Score". Suppose we are interested in the population level average of this variable. Let's call it μ. Suppose a researcher claims that the **average score is higher than 73**. Taking $\mu = 73$ as the null hypothesis, check the researchers claim. What conclusion do you make? (use 0.05 as the cut off for p-value)

iii. Construct a 95% confidence interval for the population average. Use the confidence interval to test $H_0 : \mu = 73$ vs $H_a : \mu \neq 73$.

Questions 2:

i. Calculate the proportion of building with a score more than 73.

ii. Check if the population level proportion of buildings with score more than 73 is 0.5 or not. (use 0.05 as the cutoff for p-value)

Instructions for completing this worksheet:

- Create a R markdown file, use your name and student id as the author name.
- Use the command '\pagebreak' between your answers to Q1 and Q2 (this will make sure when you convert to pdf, answer to Q2 starts on a new page)
- Create an html output first, open it in any browser and then print or save as a pdf. (if you already know how to create a pdf directly from R studio then disregard this step)
- Upload your pdf to crowdmark in the place designated for Q1.
- Drag the pages corresponding to Q2 to the place designated to Q2.
- Finish submitting everything by 6pm please.